

The growing risk of cycling throughout the Netherlands.

Group 23

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1 Introduction

The chosen article (news article) comes from a renowned news paper *Algemeen Dagblad*, translated for the *Daily newspaper*, based in Rotterdam. The article addresses the findings of the research done by the RIVM [1].

This research investigates the increasing number of accidents involving cyclists and hikers. The *Algemeen Dagblad* article was written by Martijn Klerks and published on 30 July 2026 making this a recent problem.



Figure 1: Screenshot of the article headline.

1.1 Problem statement

Cycling is rooted in Dutch culture and is the main mean of transportation of most citizens. With this in mind the record high of reported unilateral accident in 2025 involving bikes poses a huge issue on bike safety. According to the article, factors at fault might be the increasing amount of vehicles on the roads, the aging population and weather conditions too. Hence, we are aiming to figure out *how can behavioral, vehicle, and infrastructural factors be targeted to reduce the rising rate of serious single-vehicle e-bike and bicycle accidents in the Netherlands?*

1.2 Motivation

We are students living in the Netherlands whose main method of transportation is cycling. Seeing an increasing trend in bike accidents influences our perception of road safety negatively.

1.3 Stakeholders

Bicycle road safety is everybody's concern. From the individuals traveling on a bike (especially children, students, and the elderly) who can suffer life altering injuries during an accident on the roads, to municipalities and the government whose job is to maintain and increase public safety. Furthermore, car drivers also have stakes in this issues, because getting themselves involved with a bike accident is highly undesirable for them.

2 Group member 1

I chose an original research article[6] from the Accident Analysis & Prevention journal. The paper investigates how cycling accidents had increased over the years up to 2022. It highlights that, although the Safe System principles had been successfully implemented for road safety and have reduced the total number of road deaths, cycling accidents have been increasing. Cyclists had accounted for a growing share of road deaths and serious injuries. The paper's writers question why the Safe System principles had not contributed to the reduction of accidents among cyclists by studying cyclists' behaviour and accident circumstances among other factors.

3 Group member 2

To understand bike safety, we must also understand how different *species* of bikes impact trauma in a potential accident. *E-bikers at risk for severe traumatic brain injury and skull fractures by W.R. van der Spek et al.*[4], a retrospective study article, investigates the relationship between e-bike usage and severe brain injury at accident take place between 2018 and 2022 in a cross sectional

analysis. The authors state that e-bike usage is "an independent predictor for sever traumatic brain injury following a bicycle crash".

4 Group member 3

My paper is an original article in the Traffic safety journal. [3] This article focuses on understanding the underlying causes of bicycle accidents, taking data from different sources, along with showing how in the recent years the accident count increased. Using this paper we can find the common causes of the accidents, aiming to find a variable that correlates over the years with the increase of accidents.

5 Group member 4

The selected paper [5] is a systematic review that analyzes risky cycling behaviors worldwide. The authors divide these behaviors into four key dimensions:

- Lack of alertness
- Lateral instability
- Lack of protective equipment and conspicuity aids
- Errors and violations

The added value of this paper lies in this four-dimensional framework. It helps identify the underlying behavioral and environmental causes of cycling accidents, providing key insights into the recent rise in single-vehicle crashes said by *Algemeen Dagblad*.

6 Translated newspaper

The following translations is made by gen ai, specifically Gemini 3.8 Flash.

Accident figures show a worrying trend among cyclists and motorcyclists: significantly more often involved in single-vehicle accidents

The police are recording a growing number of single-vehicle accidents in which cyclists are injured, or worse. Last year, this happened an average of seven times a day: more often than with motorists.

A cyclist losing control of their handlebars, and another who can no longer avoid a lamppost: it is happening more and more often, and increasingly with more than just material damage.

Last year, the police recorded nearly 3,100 single-vehicle accidents involving cyclists—accidents where no other road user was involved. In over 2,400 cases, the cyclist was injured in those accidents, or worse.

The latter is not only a record number of bicycle accidents, but it is also the first time that cyclists have taken the 'top spot' regarding the number of serious single-vehicle accidents, replacing motorists.

Total number of accidents decreased

And these figures, published this week by Rijkswaterstaat (the Dutch

national infrastructure agency), are just the tip of the iceberg. Because the police do not show up to every accident by a long shot. Previous research in the province of Utrecht showed that the number of victims increases sixfold if you also include ambulance figures. These figures are not yet available for the Netherlands as a whole.

In total, the police recorded over 22,000 accidents last year in which no other road users were involved, over 50 percent more than in 2024. Moreover, these accidents ended more often with physical injuries. Extra striking: the total number of accidents has actually fallen for the second year in a row.

Besides cyclists, there are many motorists among the victims, but also increasingly more motorcyclists and e-bike riders. In addition to an average of six injured cyclists, the police also have to deal daily with two motorcyclists who do not escape a single-vehicle accident unscathed, and one traffic victim on an electric bicycle.

The elderly and fatbikes

The rising number of accidents involving (electric) cyclists and motorcyclists is increasing for two reasons. To begin with, the number of vehicles has increased: the number of motorcycles is now approaching 750,000, about 50,000 more than barely three years ago. Partly due to the aging population, (electric) cycling is on the rise. An organization like Veilig Verkeer Nederland (Safe Traffic Netherlands) has argued for some time that we should wear helmets and that roads should become more bicycle-friendly.

Traffic psychologist Adriaan Heino points not only to more elderly people in traffic due to the aging population, but also to more fatbikers: "The elderly fall more often because they cognitively decline, and they do so on an e-bike that rides very differently than what they are used to, while they are physically more vulnerable. Young people on fatbikes often have less of an understanding of the risk they are taking."

Record on 60-kilometer roads

In addition, the weather plays a role in the increase in the number of accidents: the summer of 2025 was warm, but despite two heat

waves, not too warm, and it also remained exceptionally dry. Perfect weather, therefore, to go out on an electric, motor, or normal bicycle.

You can also see this in the locations where the accidents happen: still most often within built-up areas (city limits), but a record number of single-vehicle accidents has been registered on 60-kilometer (37 mph) roads.

Nevertheless, it is mainly the extra bicycle accidents, even within built-up areas, that worry Heino. "You see that the number of accidents is rising faster than the number of cycled kilometers." Roads that are better designed for bicycle traffic, or bicycle paths with a speed limit, like the ones the Utrecht town of Houten is currently experimenting with, the traffic psychologist calls 'an interesting idea'. "For decades we have tailored all our roads to cars, but why should you still do that?"

Martijn Klerks; July 30, 2026, 18:00; Last update: July 31, 2026, 10:42 [2]

7 References

References

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